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# Building Mobile and Wireless Solutions for Palm Powered™ devices

Rob Veitch
Director of Business Development

PalmSource
Conference 2000

# iAnywhere Solutions, Inc.

- World leader in mobile computing
  - A decade of experience in e-Business Anywhere
  - Mobile database market leader (61% market share)
- M-Business solutions and services
  - Over 500 application partners
  - Over 10,000 corporate developers
- M-Business Platform
  - *Always Available* solutions
  - Data management and transactional synchronization
  - Wireless application server



# Vision

- The value in mobile computing is in extending the enterprise
  - Customer ownership
  - Productivity gains



# Financial Services

- Sybase technology powers ninety percent of the world's securities firms and sixty percent of its banks
- Mobile financial service solutions include
  - Stock trading anywhere
  - Agent and branch office automation
  - Remote banking
  - Kiosk systems
  - Mobile portfolio management



# Customer Examples Financial Services
# Prudential Insurance and EZ Data











- Solution
  - Sales force automation
  - Built by EZ Data
  - Deployed to Prudential Insurance Sales Force
  - 12,000 seats across 300 offices
- Benefits
  - Field agents can access customer information anywhere, anytime
  - Significant increase in employee productivity and customer service

# Healthcare

- Top 5 U.S. managed care organizations run on Sybase
- Over 450 healthcare organizations use Sybase
- Point of Care Solutions
  - Mobile care delivery
  - Physician's orders and results such as drug administration,prescription transmission, lab orders and results tracking
  - Intelligent devices used for diagnostics, patient monitoring



# Customer Examples at the Point-of-Care iScribe



- Solution
  - Prescription writing
  - Physicians use wireless PDA
  - Linked to insurance, pharmacy
- Benefits
  - Access to accurate information and online update at point of care
  - Reduced errors at pharmacy
  - Better guideline compliance
  - Reduction in overall cost

# Field Force Automation

- De facto standard for mobile sales force automation

- Radnet Inc.
- Dendrite International
- Answerthink
- IMS Health Strategic Tech.
  Gateway Systems
- PeopleSoft (Vantive)
- Onyx Software

- Pivotal Software
- Network Associates
- FDM Software
- Pacific Scientific Instruments
- Sequoia Software
- Siebel Systems, Inc.



# iAnywhere TimeTracker

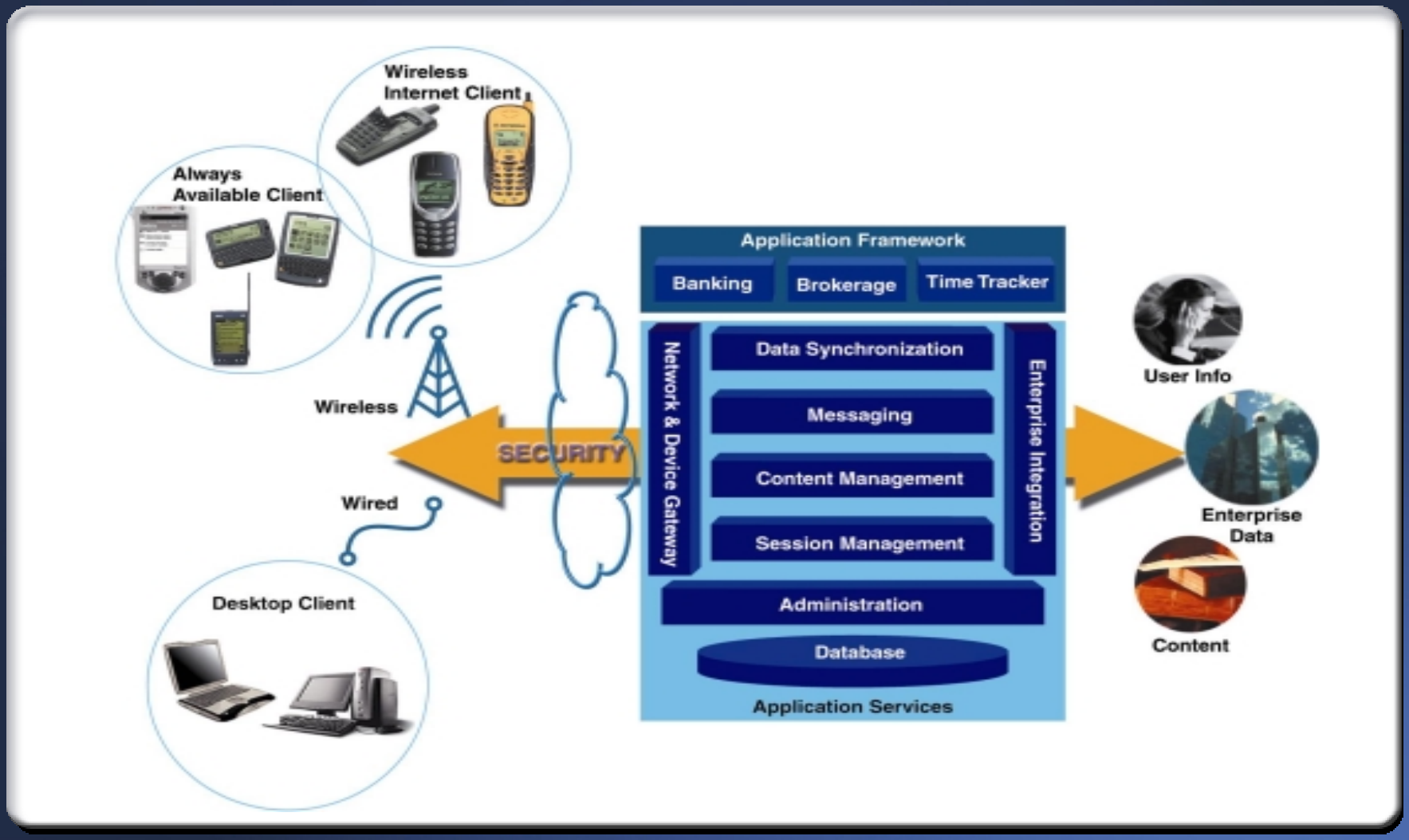
- Mobile Time & Expense Reporting
  - Capture / review data on PDA
  - Synchronize directly into SAP R/3

- Benefits
  - Reduce time between service and billing
  - Reduced errors
  - Reduced administration costs




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# m-Business Platform



# M-Business Platform Requirements

- ‘Always Available’ Access to Information
  - Mobile workers have access to information at all times and can be alerted and take action immediately
- Multi-device and Network Support
  - Easily retarget enterprise data to multiple devices on multiple networks
- Open Enterprise Integration
  - Applications can connect to a variety of back office systems
- Reliability, Availability, Scalability
  - A solid foundation that can handle peak loads and future growth
- Security
  - An enterprise can be sure that data is kept safe and secure at all times

# Always Available

- The four horsemen of wireless
  - Cost
  - Coverage
  - Battery life
  - Bandwidth
- Always Available
  - Multiple access methods: wired, wireless, voice
  - Offline transactions & transactional synchronization
  - Messaging

# Always Available Local Data Storage

- Requirement for Some Form of Local Data
  - Full-featured or lightweight
- Small Footprint, Full Featured Database
  - Adaptive Server Anywhere
- Customizable for Handhelds
  - UltraLite deployment option
  - Ideal for lightweight deployments
  - Application-optimized footprint as small as 50K

# Always Available Enterprise Data Synchronization

- MobiLink Synchronization Technology
  - Scalable, bidirectional, transactional synchronization
  - Heterogeneous synchronization
  - Utilizes standard Internet and wireless protocols
- Advanced synchronization features
  - Conflict detection and resolution
  - Support for subsetting of data
- SQL script based
  - Only write scripts for events that are needed
  - Extremely flexible

# Always Available Enterprise Data Synchronization

- Consolidated database support
  - Sybase Adaptive Server Anywhere
  - Sybase Adaptive Server Enterprise
  - Oracle 8i
  - Microsoft SQL Server 7
  - IBM DB2 UDB 5.2
- Scalability
  - Connection Pooling
  - Script Caching
  - Proven with thousands of simultaneous users
  - Performance whitepaper on the Sybase Developers Network

# Always Available Enterprise Data Synchronization

- Server-Initiated Synchronization
  - Data synchronization be initiated by enterprise events
  - Suitable if updated information on enterprise server is essential for the mobile worker
- Different devices have varying functionality
  - E.g., Symbol SPT1740s and the Qualcomm pdQ can support server-initiated synchronization
  - Other candidates include Novatel Minstrel and OmniSky support

# Always Available Messaging

- Message queuing and delivery
  - Standards-based messaging
    - JMS-based server-side messaging API
  - Store and forward message queuing with guaranteed delivery
  - Network and device independence (roaming)
  - Intelligent (program to program) and simple (email, SMS, paging)
- Network and device support
  - ARDIS, Mobitex, CDPD, DataTAC, NORCOM, Analog cellular, GSM, CDMA, Wireless LAN
  - Palm, CE, DOS, Windows, EPOC, RIM, phone
  - Paging networks
  - HDML alerts, email, SMS

# Multi-Device and Network Support Device Appropriate Content

- Device and Browser Identification
  - Notebook/desktop
  - Handheld
  - Phones
- Ability to Customize Content and Distribution
  - Based on differing devices, user preferences, networks
  - Serve up device appropriate content (HTML, WML, etc.)
  - Translate enterprise data to XML and transform to other markup languages using XSL transformations (XSLT)
  - Java-based application framework to process requests and generate content

# Multi-Device and Network Support iAnywhere Framework

- Servlet library for generation of dynamic content
  - Laptops to PDAs to phones to pagers
  - Java servlets, XML/XSLT, JSP, Dynamic XML (Cocoon)
  - HTML, HDML, WML, CHTML, PQA
- Simplified application programming
  - Automatic user identification
  - Sophisticated user authorization
  - Session management and data access
  - Filter/modify/cache requests and responses
  - Pre-built user interface (e.g., login screens)
- Integration with existing Web environment
  - Firewalls, Web servers

# Multi-Device and Network Support Other Features

- Messaging over Multiple Networks
  - Network and device independence enables roaming between networks
  - Consistent API across all platforms and networks
  - ARDIS, Mobitex, CDPD, DataTAC, Wireline, Analog cellular, GSM, CDMA, LAN, Wireless LAN, NORCOM, EDACS
- IP over Wireless Networks
  - IP compatibility for non-IP networks
  - Increase speed and reduce data
  - Packet:  ARDIS, Mobitex, CDPD, DataTAC
  - Circuit switched:  GSM, CDMA, DoPa, LAN, GPRS, PDC, PHS, Nextel, and Wireless LAN
- Voice Services
  - VoiceXML

# Open Enterprise Integration Data, Applications, Messages

- Database Integration
  - Open database support for all leading enterprise vendors including Microsoft, Oracle, IBM, Sybase
  - ODBC, JDBC (native for Java components), Sybase Open Client, Oracle OCI
- Application Integration
  - Provides component based access to ERP (SAP), CRM, mainframe, legacy
- Messaging
  - Integration with existing messaging systems
  - Advanced messaging services through TIBCO, MQ Series, and other XML-based systems

# Open Enterprise Integration Standards

- XML support
  - Use SQL to query any database (Sybase, Oracle, IBM, etc.) and return an XML document
  - Update data in a SQL database based on an XML document using pre-defined DTD for updates
  - XML-based messaging systems
- Open component model for custom logic
  - Enterprise JavaBeans (EJB), CORBA, C++
  - Build components to support new applications
  - Use existing skills for faster development cycles
- Extensive Java 2 Enterprise Edition (J2EE) support
  - EJB, Servlets, JSP, JTS/JTA, JNDI

# iAnywhere Wireless Server Reliability, Availability, Scalability

- Load Balancing Extremely Flexible and Robust
  - Client load can be distributed over multiple servers
  - Flexible load distribution: random, round robin, weighted, dynamic
- Failover Support for High Availability
  - Latency almost undetectable since fail over information local
- High Performance
  - Instance pooling, thread pooling, and efficient HTTP session management
  - Database connection pooling
  - Proven to handle millions of transactions per day with thousands of concurrent users
- Transaction Management
  - CORBA (OTS), JTS, Microsoft DTC

# iAnywhere Wireless Server End-To-End Security

- Integration With Existing Security Systems
  - Open, standards-based approach
  - Provides for integration of firewall and Virtual Private Network (VPN) technology
  - Supports PKI
- Authentication, Authorization, Encryption
  - Flexible options for user authentication
  - Several options for role-based security
  - Simplifies adding SSL support to your applications
  - Security management tools

# The Portal as Single Point of Contact




*Web Browsers*



*Mobile Devices*

**Enterprise Portal**

**Sales** **Customer Service** **Manufacturing**
**Marketing** **Shipping**

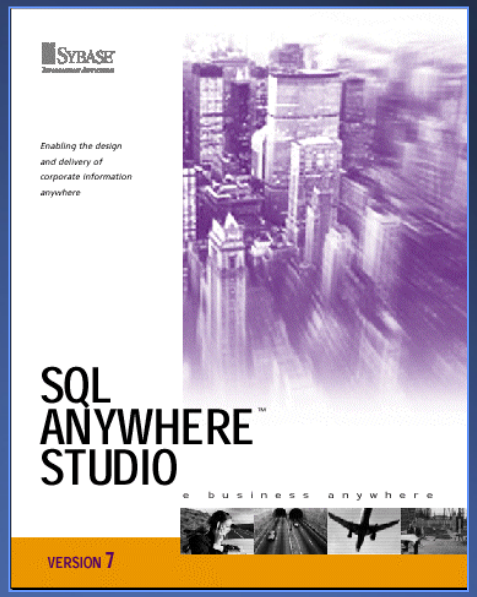
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# Products & Services

- SQL Anywhere Studio
  - Mobile & workgroup DBMS & transactional synchronization
- iAnywhere Wireless Server
  - Mobile & wireless platform supporting Always Available applications
- iAnywhere TimeTracker for SAP R/3
  - Mobile time reporting for SAP CATS
- iAnywhere Services
  - Mobile & wireless application architecture, design, and implementation services



# iAnywhere Solutions & Palm Technical Seminar Series

Learn who to build and deploy secure, scalable mobile & wireless solutions

| | |
|---|---|
| – Atlanta, GA | January 30th |
| – Durham, NC | February 8th |
| – Austin, TX | February 15th |
| – San Francisco, CA | February 21st |
| – San Diego, CA | March 6th |

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